FINANCIAL RISK AND TRADING SYSTEM – PROJECT SUMMARY

Introduction: The Financial Risk and Trading System (RPAS) is designed to automate portfolio risk assessment, backtesting of trading strategies, portfolio optimization, and transaction cost analysis. It provides financial analysts and portfolio managers with a unified, automated, and audit-ready platform that ensures accurate, consistent, and efficient decision-making in volatile markets.

Aim of the Project: The project aims to design and develop a comprehensive system that integrates Risk Analytics, Backtesting, Portfolio Optimization, Transaction Cost Analysis (TCA), and Reporting into a single framework to improve transparency, performance, and compliance in portfolio management.

Objectives: 1. Automate risk measurement using metrics like Value at Risk (VaR) and Expected Shortfall (ES). 2. Enable backtesting of trading strategies under realistic market conditions. 3. Optimize portfolio allocation using quantitative models. 4. Evaluate and reduce transaction costs. 5. Generate comprehensive, audit-ready reports for decision-making.

Scope: The system focuses on portfolio analytics, excluding live trading and tax reporting. It covers data ingestion, risk evaluation, backtesting, optimization, TCA, and reporting.

Existing System: In current practices, different tools are used for risk, optimization, and reporting, often leading to manual errors and inefficiencies. The absence of automation, integration, and real-time insights limits performance and increases operational risks.

Proposed System: The proposed RPAS replaces fragmented systems with an integrated, automated platform that performs all analytical tasks efficiently. It provides real-time insights, reproducible results, and audit trails. Key modules include: - **Risk Management:** Calculates VaR, ES, and stress test metrics. - **Backtesting:** Simulates historical strategies including momentum and mean reversion. - **Portfolio Optimization:** Identifies efficient portfolios balancing risk and return. - **Transaction Cost Analysis:** Evaluates execution quality and cost efficiency. - **Reporting:** Produces interactive dashboards and detailed reports.

System Architecture: The architecture consists of five layers: 1. External Data Sources – Market feeds, trade logs, and portfolios. 2. Data Ingestion & Validation – Cleans and verifies data. 3. Database & Storage – Stores validated data and analytical outputs. 4. Processing Modules – Runs analytics like risk, optimization, backtesting, and TCA. 5. Reporting & Dashboards – Delivers visual insights and performance metrics.

Daily Risk Workflow: The workflow automatically loads data, validates it, computes VaR and ES, runs stress tests, and generates alerts and reports before market hours. This ensures timely and accurate risk updates.

Technology Stack: Python (NumPy, Pandas, Matplotlib), Power Query, MySQL Workbench, scikit-learn, and Excel are used for data processing, analytics, and reporting. Security is ensured through encryption and role-based access.

Modules Overview: - **Data Ingestion & Validation:** Ensures clean, accurate data input. - **Database Management:** Maintains structured storage with version control. - **Risk Analytics:** Measures VaR, ES, volatility, and performs stress testing. - **Backtesting:** Tests trading models for consistency and robustness. - **Portfolio Optimization:** Uses Modern Portfolio Theory (MPT) for efficient allocation. - **Transaction Cost Analysis:** Analyzes slippage, commissions, and price impact. - **Reporting & Dashboards:** Generates reports in Excel, PDF, and dashboards. - **Security & Access Control:** Manages roles, authentication, and encryption.

Outputs and Results: 1. **Risk Analytics:** Produces VaR, ES, cumulative returns, and drawdown graphs. 2. **Backtesting:** Evaluates strategy performance (Buy & Hold, Momentum, Mean Reversion). 3. **Portfolio Optimization:** Displays efficient frontier and optimal weights. 4. **TCA:** Shows execution cost, price impact, and shortfall analysis.

Testing and Validation: Includes unit, integration, system, and performance testing. Validation ensures correctness of calculations, workflow integrity, and report accuracy. UAT ensures user satisfaction and business requirement alignment.

Deployment and Maintenance: The system supports cloud or on-premises deployment. Backup, recovery, monitoring, and updates ensure long-term reliability. Security measures include encryption and audit logging.

Future Enhancements: - Real-time intraday risk monitoring. - Support for more asset classes. - Integration of AI/ML for prediction and optimization. - Advanced visualization and cloud scalability.

Conclusion: The Financial Risk and Trading System is a complete, automated, and modular platform that strengthens financial risk analysis, trading strategy validation, and portfolio management. It improves transparency, reduces manual effort, and supports data-driven investment decisions in an evolving financial landscape.